PROJECT PROPOSAL

SmartDine – Restaurant Ordering & Management System

1. Project Name:

SmartDine – Restaurant Ordering & Management System

2. Student Name:

Muhammad Zain Tahir

3. Roll Number:

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4. Start Date:

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5. Project Description:

SmartDine is a modern web-based Restaurant Ordering and Management System developed as a Final Year CIT Project. The main purpose of this project is to digitalize restaurant operations, including order management, billing, menu management, and kitchen order processing.

The system will allow restaurant staff to manage dine-in, takeaway, and delivery orders efficiently. It will provide a digital billing system, help track order status, and improve communication between the cashier and kitchen staff.

The project will be developed using the MERN Stack to provide a responsive, user-friendly, and efficient restaurant management solution.

6. Technology Stack / Tools:

- **MongoDB:** Database for storing menu items, orders, users, bills, and other restaurant data.
- **Express.js:** Backend framework for developing APIs and handling application requests.
- **React.js:** Frontend library for building an interactive and responsive user interface.
- **Node.js:** JavaScript runtime for running the backend server.
- **HTML5:** Structure of web pages.
- **CSS3:** Styling and responsive layouts.
- **JavaScript:** Application logic and interactive functionality.
- **Bootstrap:** Responsive design and user interface components.

- **Visual Studio Code:** Code editor for project development.
- **Git & GitHub:** Version control and project code management.
- **Postman:** Testing backend APIs.
- **npm:** Managing project dependencies and packages.

7. Problem to Be Solved:

Many restaurants manage orders and billing manually, which can cause calculation errors, delays, and confusion between cashiers and kitchen staff. Manual record-keeping also makes it difficult to track order progress and maintain accurate sales records.

SmartDine will solve these problems by providing a centralized digital system for managing restaurant orders, bills, menu items, and kitchen workflow. It will reduce manual work, improve billing accuracy, and make restaurant operations more organized and efficient.

8. Project Duration:

Total Duration: 6 Months

9. Weekly Modules / Development Plan:

Module 1: Project Planning and Requirements Analysis (Weeks 1–2)

- Identify project objectives and requirements.
- Analyze restaurant ordering and billing workflows.
- Define system features and user roles.
- Prepare the project structure and development plan.

Module 2: UI/UX Design and Frontend Setup (Weeks 3–4)

- Design the SmartDine user interface.
- Create responsive layouts using React.js, CSS3, and Bootstrap.
- Develop the navigation system and dashboard layout.
- Establish the frontend project structure.

Module 3: MERN Stack Environment Setup (Weeks 5–6)

- Set up Node.js and Express.js.
- Configure the React.js application.
- Connect the application to MongoDB.
- Organize frontend and backend folders.

Module 4: Database Design and Authentication (Weeks 7–8)

- Design MongoDB collections for users, menu items, and orders.

- Develop user registration and login functionality.
- Implement role-based access for authorized users.
- Configure backend validation and error handling.

Module 5: Menu Management System (Weeks 9–10)

- Add, update, and delete menu items.
- Organize items by categories.
- Display item names, prices, and availability.
- Connect menu management with MongoDB.

Module 6: Order Management System (Weeks 11–12)

- Create new restaurant orders.
- Support dine-in, takeaway, and delivery orders.
- Add or remove items before order confirmation.
- Store orders and retrieve order details through APIs.

Module 7: Billing and Invoice System (Weeks 13–14)

- Generate bills automatically from selected items.
- Calculate totals and payments accurately.
- Handle received amounts and remaining balances.
- Provide a printable invoice or receipt.

Module 8: Kitchen Order Processing (Weeks 15–16)

- Send confirmed orders to the kitchen order dashboard.
- Display ordered items and order details.
- Track order status: Pending, Confirmed, Preparing, Ready, and Completed.
- Allow staff to update order progress.

Module 9: Table, Takeaway, and Delivery Management (Weeks 17–18)

- Manage dine-in table orders.
- Organize takeaway orders.
- Record delivery details where required.
- Track order types and their statuses.

Module 10: Dashboard and Sales Reports (Weeks 19–20)

- Display order and sales summaries.
- Show daily sales and order statistics.
- Maintain transaction and order records.
- Provide basic reports for restaurant management.

Module 11: Testing and Bug Fixing (Weeks 21–22)

- Test frontend and backend functionality.
- Test API endpoints and database operations.
- Identify and fix errors.
- Verify billing calculations and order status updates.

Module 12: Final Integration and Documentation (Weeks 23–24)

- Integrate and review all project modules.
- Improve responsiveness and usability.
- Prepare project documentation and screenshots.
- Prepare the final project demonstration and presentation.

Expected Outcome:

The expected outcome is a functional, responsive, and user-friendly restaurant management web application built using the MERN Stack. SmartDine will help restaurant staff manage orders, generate bills, monitor kitchen preparation, and organize restaurant operations through a centralized system.

Submitted By:

Muhammad Zain Tahir

Project Title:

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